# KARTHICK CHIDAMBARAM SUNDARESAN

**Senior Backend & Microservices Engineer | Lead Software Developer**

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## EXECUTIVE SUMMARY

Versatile Senior Software Engineer & Technical Lead with 8+ years of enterprise software engineering experience overall, including close to 4 years based in Singapore delivering mission-critical systems for the country's financial and public-sector institutions. Currently building derivatives settlement and market-data infrastructure for Singapore Exchange (SGX), and previously led core banking microservices for OCBC Bank and a government contractor portal for the Land Transport Authority (LTA). Deep technical expertise in Java (8-25), Python, Spring Boot, Apache Kafka, Microservices Architecture, and Cloud Engineering (GCP, AWS, Azure), complemented by applied AI/LLM engineering (Spring AI, RAG, Anthropic Claude & OpenAI APIs) to accelerate delivery velocity. Proven track record of leading end-to-end backend development, modernizing legacy systems, and integrating quantitative derivatives libraries and enterprise-grade security across tier-1 banking, exchange, and government clients.

## CORE TECHNICAL CAPABILITIES

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|---|---|
| **Backend & Core** | Java (8-25), Python, Spring Boot, Spring Batch, Spring Data JPA, RESTful APIs, Microservices, Design Patterns (Saga, CQRS, Database-per-service, Circuit Breaker, API Gateway) |
| **AI & LLM Engineering** | Spring AI, LangChain4j, Retrieval-Augmented Generation (RAG), Vector Databases (pgvector, Pinecone), Anthropic Claude API, OpenAI API, Model Context Protocol (MCP), Prompt Engineering |
| **Banking & Payments Domain** | Derivatives Options Settlement, SWIFT MT (MT101, MT103, MT202, MT940/942), ISO 20022 MX (pacs.008, pain.001, camt.053), Multi-Currency Fixed Deposits, Virtual Accounts (VAM), Interbank P2P Payments |
| **Cloud & DevOps** | GCP (Cloud Run, GCS, CloudSQL, BigQuery), AWS (EC2, IAM), Azure (App Services), Docker, Red Hat OpenShift, Kubernetes, CI/CD (GitHub Actions, Jenkins, GitLab CI), SonarQube |
| **Data & Streaming** | Apache Kafka, Prefect Orchestration, Parquet Data Lake, Pandas, NumPy, SciPy, MySQL, PostgreSQL, Oracle 12c, MongoDB, Redis |
| **Engineering Tools & Practices** | AI-Assisted Engineering (Claude, Microsoft Copilot), Git, Bitbucket, Swagger, Kibana, JIRA, Confluence, Agile, Security (OAuth 2.0, JWT, RBAC), UNIX Shell Scripting |
| **Front End & Scripting** | HTML5, CSS, JavaScript, JQuery, JSP, ReactJS, PEGA, Deluge Scripting - Zoho, Unix Shell Scripting |

## PROFESSIONAL EXPERIENCE

### Experienced Consultant - Lead Software Engineer — Jan 2026 - Present

*Amaris | Client: Singapore Exchange (SGX) - Derivatives Engineering | Singapore*

- **Automated Derivatives Ingestion Pipeline:** Architected and delivered core backend ingestion and surface-building pipelines using Python, Prefect, and GCP (Cloud Run, GCS) to automate daily settlement price (DSP) processing for thousands of JPX/SGX options contracts.
- **Data Pipeline & Mathematical Library Integration:** Integrated Python data orchestration with financial/scientific libraries (tf_quant_finance, SciPy) to execute Black-76 valuation algorithms, Newton-Raphson solvers, and no-arbitrage validations across multi-source derivatives market data streams.
- **High-Performance Analytics & Storage Engine:** Built automated Master Market Data pipelines parsing panelist files (IMC, Super Derivatives) and exchange reports, persisting processed analytics into optimized Parquet columnar files on GCP and PostgreSQL for audit-ready web applications.
- **AI-Assisted Engineering & Velocity:** Leveraged AI-assisted development tools (Claude, Microsoft Copilot) to accelerate Python pipeline development, automate PyTest coverage, and optimize complex mathematical logic execution.
- **DevOps & Reliability Optimization:** Implemented end-to-end automated validations across daily derivatives clearing workflows and spearheaded repository/pipeline migration from Bitbucket/Jenkins to GitHub Actions and GCP Cloud Run for zero-downtime container deployments.

### Senior Software Engineer - Microservices — Nov 2023 - Jun 2025

*Persolkelly (Contract) | Client: OCBC Bank - Group Operations & Technology | Singapore*

- **Multi-Currency Core Banking Engine:** Engineered core backend microservices (Placement, Withdrawal, Change Maturity) for Local (LCY) and Foreign Currency (FCY) Time Deposits using Java 21, Spring Boot, and REST APIs integrated with Silverlake Axis Core Banking (SIBS) and PEGA frontends.
- **Java 21 Upgrade & System Refactoring:** Led backend legacy framework upgrades from Java 8 to Java 21 across core modules, refactoring dependent SOA libraries and ensuring microservice integration compatibility.
- **Batch Rate & Enterprise Integration Engine:** Developed Spring Batch services to ingest and calculate dynamic branch, penalty, and empowered interest rates; integrated REST/MQ interfaces with Silverlake Card System for customer data synchronization.
- **Core Transaction Performance Optimization:** Refactored multi-currency transaction processing modules, reducing database lock contention and achieving a 35% improvement in peak-hour batch execution speed.
- **OpenShift Cloud Deployment & Rewards Replatforming:** Integrated regional SOA services across SG, MY, and HK for credit card admin processing (KrisFlyer, Shopee) and deployed containerized services to Red Hat OpenShift clusters via JIRA-driven CI/CD pipelines across SIT, UAT, and PROD while mentoring junior developers.

### Full Stack Developer — Nov 2022 - Nov 2023

*Technopals -> Wipro (Contract) | Client: Land Transport Authority (LTA) | Singapore*

- **Enterprise Portal & API Development:** Engineered core features of the Document Approval & Receipt (DAR) contractor portal using ReactJS, Spring Boot, and MongoDB with Azure AD authentication, enabling multi-gigabyte file handling (up to 1 GB per file).
- **Automated Data Transfer & SLA Optimization:** Authored Python automation scripts to route files from cloud landing zones to on-premise Documentum repositories, reducing production processing incidents by 90% and maintaining compliance with government SLAs.
- **Workflow Automation & Access Control:** Implemented Role-Based Access Control (RBAC) and Spring Batch validation workflows on Oracle DB to automate acceptance/rejection PDF generation for project stakeholders.
- Built file upload, validation and DAR approval workflows in ReactJS, with robust error handling and MongoDB based user authentication for contractors.

- **Secure Cloud Integration & Microservices Security:** Applied OAuth 2.0 and JWT token authentication mechanisms across public and internal API endpoints to enforce stringent government data privacy guidelines.
- **Infrastructure Maintenance & Purge Automation:** Monitored and maintained Azure App Services ensuring high availability, and developed automated Python purge scripts for archived DAR files, significantly improving batch job efficiency.

### Senior Software Engineer — Oct 2021 - Aug 2022

*Larsen & Toubro Infotech (LTI) | Client: Citi Bank (US) | Chennai, India*

- **Automated Data Screening Engine:** Designed and developed 70% of core modules for the Data Screening Tool (DST) using Spring Boot, Java 11, and Oracle DB to detect payment lifecycle data discrepancies across heterogeneous enterprise datasets.
- **Batch Reconciliation Automation:** Built automated Spring Batch jobs to identify JSON-based payload differences in Oracle DB, eliminating manual reconciliation effort by 80% and improving overall data accuracy.
- **Database Query Optimization:** Optimized complex SQL stored procedures and index structures on Oracle DB 12c, cutting nightly batch processing runtime by 40%.
- **Cloud Deployment & Quality Standards:** Integrated SonarQube to eliminate code vulnerabilities and code smells, deploying microservices to AWS EC2 under strict DevOps compliance protocols for SIT, UAT, and Production.

### Application Development Senior Analyst — Mar 2021 - Oct 2021

*Accenture | Client: Standard Chartered Bank (UK) | Chennai, India*

- **Real-Time Payment Engine Development:** Spearheaded backend microservices development (60% core ownership) for Standard Chartered Pay (SCPAY), enabling 24/7 interbank peer-to-peer (P2P) and merchant payments via mobile, email, or FPS ID without account number dependency.
- **Event-Driven Kafka Architecture:** Architected real-time asynchronous microservices using Apache Kafka to ensure high-throughput, fault-tolerant communication across user onboarding, core banking integration, accounting, and notification modules.
- **Resilient Transaction Fault Handling:** Implemented retry mechanisms and dead-letter queue (DLQ) patterns in Spring Kafka to ensure zero transaction loss during peak payment surges.
- **Microservices Security & Agile Governance:** Integrated OAuth 2.0 API Gateway security policies and rate-limiting to protect financial endpoints while managing feature stories on JIRA in a fast-paced Agile setting.

### Software Developer (Gulf Corporate Banking Specialist) — Jun 2018 - Mar 2021

*Mindgate Solutions | Clients: Mashreq Bank (Dubai), Gulf International Bank (GIB), FAB | Chennai, India*

- **Corporate Payment & Authorization Engine (Gulf Market):** Built corporate banking payment modules for single/bulk transfers supporting SWIFT MT messages (MT101, MT103, MT202, MT940/942) and ISO 20022 MX standards (pacs.008, pain.001, camt.053), implementing corporate group limit matrix validation logic.
- **Monolith to Microservices Migration:** Migrated 80% of legacy J2EE monolithic customer onboarding and Virtual Account Management (VAM) systems to modern Spring Boot microservices on Azure utilizing Service Discovery, Config Server, and API Gateway patterns.
- **Liquidity Management Module:** Developed a Virtual Account hierarchy budgeting module for Mashreq Bank Dubai, enhancing multi-entity fund allocation for regional corporate clients across UAE and GCC.
- **High-Throughput ISO 20022 Parser:** Engineered custom XML/JSON parsing engines in Java to validate complex ISO 20022 payment payloads against stringent regional clearing rules.
- **Bulk Processing & Validation Engine:** Built Spring Batch services for bulk payment processing (CSV/Excel) with validations via stored procedures for own account and international cross-border transfers.

### Associate Software Engineer — Oct 2017 - Mar 2018

*Dot2Globe Solutions | Client: SkyOnline (China) | Chennai, India*

- **ERP & Marketplace API Integrations:** Built third-party marketplace API integrations (Amazon, Shopify, eBay) with OAuth 2.0 authentication flows for token lifecycle management, order processing, and inventory synchronization.
- **Inventory & Order Synchronization Engine:** Developed automated scheduled sync jobs using Java to reconcile stock levels and cross-platform multi-channel customer orders.
- **RESTful API Service Layer:** Designed modular REST controllers and data access objects (DAOs) using Hibernate/JPA to streamline backend catalog maintenance.
- **Unit Testing & Code Coverage:** Wrote comprehensive JUnit and Mockito test suites to achieve 85%+ test coverage for all backend payment and order integration services.

## EDUCATION & CERTIFICATIONS

**M.Sc. in Computer Science** — Liverpool John Moores University, UK | Dec 2021 - Feb 2023

**Post Graduate Diploma in Software Development** — IIIT Bangalore, India | Sep 2020 - Oct 2021

**B.E. in Electrical & Electronics Engineering** — Anna University (Easwari Engineering College), India | Sep 2012 - Apr 2016

### Professional Certifications

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Google Cloud Certified: Associate Cloud Engineer (In Progress)
- Java Full Stack Developer Certification — Anew Tech
- Zoho Creator Certified Associate — Zoho Corp
- Claude Code 101 — Anthropic
- Claude — AI Fluency For Builders